

Kemal Baran Dursun

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EDUCATION

McGill University

Montreal, QC

Bachelor of Arts, Computer Science & Economics · Expected December 2026
Relevant coursework: Econometrics, Probability, Computation, Data Structures, Economic Statistics, Financial Instruments and Institutions.

PROJECTS

FragMance — Full-Stack Social iOS App · GitHub

Sole Engineer · Live on the App Store

React Native (Expo) · TypeScript · Node.js/Express · PostgreSQL (Prisma) · Redis · Supabase · GitHub Actions · EAS · Sentry

- Solo-designed, built, and shipped a production social iOS app for fragrance discovery — owning UX, a 58-endpoint REST backend, the React Native client, CI/CD, and security (~18K lines of TypeScript).
- Engineered OCR- and typo-resilient search over a 63,751-fragrance catalog: PostgreSQL trigram (pg_trgm) indexes plus a confidence-ranking layer (word similarity, token recall, edit distance, accent folding) power barcode scanning and free-text lookup.
- Built a content-based recommender ranking fragrances by IDF-weighted scent-note overlap scaled by brand-tier affinity, plus a multi-signal trending model over wear logs and social activity.
- Discovered and remediated a live PII exposure in a self-initiated security audit — the database's auto-generated REST API served every user's email and push token to anyone with the app's public client key; locked it down at the Postgres grant/RLS layer and added a regression test.

EconSight — Canadian Economic Intelligence Platform (live)

Python · FastAPI · XGBoost · PostgreSQL · React

- Ingested 9 macro indicators from Statistics Canada and the Bank of Canada into a PostgreSQL medallion warehouse, driving VAR and XGBoost forecasts with Monte Carlo scenario bands.
- Surfaced a composite economic health score, RAG natural-language Q&A, and one-click PDF reports.

Personalized Newspaper — AI-Curated Daily Broadsheet (live)

Next.js · TypeScript · PostgreSQL · Drizzle · Claude API

- Built a scheduled pipeline polling Finnhub, RSS feeds, and Hacker News, scoring every article for relevance and clickbait with the Claude API.
- Rendered a three-column broadsheet, deployed as a single Railway service.

AuditAP — AP Risk & Anomaly Detection (live)

Python · SQL · PostgreSQL · LLMs

- Integrated with QuickBooks and built 90-day rolling statistical baselines to flag duplicate invoices, vendor price creep, and zombie subscriptions.
- Added an LLM audit of invoice line-item descriptions to catch anomalies rule-based checks miss.

PROFESSIONAL EXPERIENCE

Duronm Textile Company — Invoicing & Collections Intern

Izmir, Turkey · Jun–Aug 2023

- Analyzed 200+ accounts-receivable transactions for payment trends, delays, and cash-flow risk.
- Built weekly Excel reports tracking overdue balances, collection performance, and working-capital indicators for management.

Advanced Information Modeling Inc — IT Intern

St. Petersburg, FL (Remote) · Jun–Aug 2024

- Monitored system availability and performance logs, documenting recurring technical issues for the team.
- Updated internal documentation and standard operating procedures to keep support references current.
- Recommended process improvements to streamline recurring support workflows.

ADDITIONAL INFORMATION

Certifications: FMVA (Financial Modeling & Valuation Analyst); CFA Level 1 Candidate

Skills: Python, TypeScript, SQL (PostgreSQL), React / React Native, Node.js, FastAPI, Next.js, Pandas / NumPy, Power BI, Excel; Risk metrics (VaR, Expected Shortfall), Statistical analysis

Languages: English (native), Turkish (native), French (intermediate)

Activities: McGill Consulting Association; Economics Student Association